

# Geometry

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## Part 1

A *cellular embedding* of a connected graph  $G = (V, E)$  on a surface  $S$  is an embedding such that every face of  $S \setminus G$  is homeomorphic to an open disk.

For a surface of genus  $g$ :

- Orientable surface:  $\chi(S) = 2 - 2g$ ,
- Non-orientable surface:  $\chi(S) = 2 - g$ .

Throughout,  $v = |V|$ ,  $e = |E|$ , and  $f$  denotes the number of faces.

### Question 1

Provide an explicit cellular embedding of the graph in Figure 1-c (the graph  $K_4$ ) on an orientable surface of genus 1.

### Solution

The graph  $K_4$  has

$$v = 4, \quad e = 6.$$

For an orientable surface of genus 1 (the torus), the Euler characteristic is

$$\chi = 2 - 2g = 0.$$

Hence, for a cellular embedding,

$$v - e + f = 0 \quad \Rightarrow \quad 4 - 6 + f = 0,$$

which gives  $f = 2$ .

We use the square model of the torus, where opposite sides are identified with the same orientation. Place the four vertices at the four corners of the square. Draw the four edges of a cycle along the boundary of the square. Draw one diagonal inside the square, and draw the second diagonal so that it wraps around the square using the edge identifications.

Due to the torus identifications, the diagonals do not cross on the surface. The complement consists of exactly two faces, each homeomorphic to a disk. Hence this embedding is cellular.

### Question 2

Provide an explicit cellular embedding of the graph of Figure 1-c on a non-orientable Klein bottle.

### Solution

The Klein bottle is a non-orientable surface of genus 2, hence

$$\chi = 2 - g = 0.$$

As in Question 1, for  $K_4$  we obtain

$$4 - 6 + f = 0 \Rightarrow f = 2.$$

Using the polygonal schema of the Klein bottle, one may embed the four vertices and draw the six edges so that one diagonal uses the orientation-reversing identification. This removes crossings in the same way as on the torus.

The resulting embedding has two faces, each a disk, hence it is cellular.

### Question 3

Let  $G$  be a simple graph cellularly embedded in an orientable or non-orientable surface of genus  $g$ . Prove that

$$g \leq e - v + 1.$$

Can the graph of Figure 1-c be cellularly embedded in a two-holed torus?

### Solution

Let  $f$  be the number of faces. Since the embedding is cellular,  $f \geq 1$ .

**Orientable case.** Using Euler's formula,

$$v - e + f = 2 - 2g \geq v - e + 1,$$

which implies

$$g \leq e - v + 1.$$

**Non-orientable case.** Similarly,

$$v - e + f = 2 - g \geq v - e + 1,$$

which again yields

$$g \leq e - v + 1.$$

For the graph  $K_4$ , we have

$$e - v + 1 = 6 - 4 + 1 = 3.$$

A two-holed torus has genus  $g = 2$ , which satisfies the inequality. Hence  $K_4$  can be cellularly embedded on a two-holed torus.

#### Question 4

Let  $G$  be a simple graph cellularly embedded on an orientable surface of genus  $g$ . Prove that

$$\left\lceil \frac{e}{6} - \frac{v}{2} + 1 \right\rceil \leq g,$$

with equality if and only if  $G$  is a triangulation.

#### Solution

Euler's formula gives

$$v - e + f = 2 - 2g,$$

hence

$$g = \frac{e - v - f + 2}{2}.$$

Since  $G$  is simple, every face has degree at least 3. By double counting edge-face incidences,

$$3f \leq 2e \quad \Rightarrow \quad f \leq \frac{2e}{3}.$$

Substituting,

$$g \geq \frac{e - v - \frac{2e}{3} + 2}{2} = \frac{e}{6} - \frac{v}{2} + 1.$$

Since  $g$  is an integer, we obtain

$$\left\lceil \frac{e}{6} - \frac{v}{2} + 1 \right\rceil \leq g.$$

Equality holds if and only if  $3f = 2e$ , that is, every face has degree 3, so  $G$  is a triangulation.

#### Question 5

Let  $K_n$  be a complete graph cellularly embedded in a surface of genus  $g$ . Prove that

$$\left\lceil \frac{(n-3)(n-4)}{12} \right\rceil \leq g.$$

#### Solution

For  $K_n$ ,

$$v = n, \quad e = \frac{n(n-1)}{2}.$$

Applying the bound from Question 4,

$$g \geq \frac{e}{6} - \frac{v}{2} + 1 = \frac{n(n-1)}{12} - \frac{n}{2} + 1 = \frac{(n-3)(n-4)}{12}.$$

Taking the ceiling yields the desired inequality.

### Question 6

Let  $G$  be a  $k$ -connected graph,  $k \geq 4$ , cellularly embedded in an orientable surface of genus  $g$ . Let  $C_3$  be a 3-cycle that is contractible but does not bound a face.

- (a) Prove that  $C_3$  is separating.
- (b) Prove that  $G$  is planar.

### Solution

(a) Since  $C_3$  is contractible, it bounds a disk  $D$  on the surface. Because  $C_3$  is not the boundary of a face, there exist edges or vertices both inside and outside  $D$ . Removing the three vertices of  $C_3$  disconnects the graph, hence  $C_3$  is a separating cycle.

(b) A separating 3-cycle contradicts 4-connectivity. Therefore such a configuration cannot exist on a surface of positive genus. Hence the surface must be the sphere, and  $G$  is planar.

### Question 7

Provide an explicit cellular embedding of  $K_5$  on a torus.

### Solution

For  $K_5$ , we have

$$v = 5, \quad e = 10.$$

On a torus,

$$v - e + f = 0 \Rightarrow 5 - 10 + f = 0 \Rightarrow f = 5.$$

Using the square model of the torus, one can draw the five vertices and route the edges so that crossings are removed using the identifications. This classical construction yields five disk faces, hence a cellular embedding.

### Question 8

Prove existence or non-existence of a cellular embedding of  $K_4$  on a non-orientable surface of genus 3.

### Solution

For a non-orientable surface of genus 3,

$$\chi = 2 - 3 = -1.$$

Euler's formula gives

$$4 - 6 + f = -1 \Rightarrow f = 1.$$

Thus the embedding must have exactly one face. Such an embedding exists by routing the edges of  $K_4$  through the three crosscaps so that all edges lie on the boundary of a single disk.

Hence a cellular embedding of  $K_4$  on a non-orientable surface of genus 3 exists.

## Part 2 – 3-Coloring Planar Triangulations

### Question 1

#### Statement

Prove that a graph is Eulerian if and only if it is connected and all its vertices have even degree.

#### Solution

( $\Rightarrow$ ) Assume that  $G$  is Eulerian. Then there exists a closed walk that traverses every edge exactly once. Every time the walk enters a vertex via an edge, it must leave via another unused edge, except for the starting/ending vertex, where the same property holds because the walk is closed. Hence each vertex is incident to an even number of edges, so all vertices have even degree. Since the walk traverses all edges, the graph is connected.

( $\Leftarrow$ ) Assume that  $G$  is connected and every vertex has even degree. We prove that  $G$  has an Eulerian circuit by induction on the number of edges. If  $G$  has no edges, the result is trivial.

Otherwise, since all degrees are even and  $G$  is connected,  $G$  contains a cycle  $C$ . Remove the edges of  $C$  from  $G$ . The remaining graph may be disconnected, but every vertex still has even degree. By induction, each connected component admits an Eulerian circuit. By inserting the cycle  $C$  at the appropriate vertex, we obtain an Eulerian circuit of  $G$ .

Thus  $G$  is Eulerian.

### Question 2

#### Statement

Let  $G$  be a loopless graph that is a triangulation of the sphere. Prove that  $G$  is 3-colorable if and only if  $G$  is Eulerian.

## Solution

We prove both implications.

### ( $\Rightarrow$ ) If $G$ is 3-colorable, then $G$ is Eulerian

Assume that  $G$  admits a proper 3-coloring of its vertices. Consider any face of  $G$ , which is a triangle. Since adjacent vertices have different colors, every triangular face uses all three colors.

Fix a vertex  $v$ . Going around  $v$ , the colors of its neighbors must alternate between the two colors different from the color of  $v$ . Hence the number of neighbors of  $v$  is even, and therefore  $\deg(v)$  is even.

Since this holds for every vertex,  $G$  is Eulerian.

### ( $\Leftarrow$ ) If $G$ is Eulerian, then $G$ is 3-colorable

Assume that  $G$  is Eulerian, i.e. every vertex has even degree. We use induction on the number of vertices.

If  $|V(G)| = 3$ , the result is trivial. Assume the result holds for all smaller Eulerian triangulations.

Since  $G$  is a triangulation, it has a vertex  $v$  of degree at least 4. Because  $\deg(v)$  is even,  $\deg(v) \geq 4$ . Let  $v_1, \dots, v_{2k}$  be the neighbors of  $v$  in cyclic order. Remove  $v$  and add edges between  $v_1, v_2, v_3, v_4, \dots, v_{2k-1}, v_{2k}$ . The resulting graph  $G'$  is again a triangulation and all degrees remain even.

By the induction hypothesis,  $G'$  is 3-colorable. Now assign to  $v$  the unique color not used by its neighbors  $v_1, \dots, v_{2k}$ . This gives a proper 3-coloring of  $G$ .

Thus  $G$  is 3-colorable.

We conclude that a planar triangulation is 3-colorable if and only if it is Eulerian.

## Part 3 – 6-Degree Theorem

### Question 1

#### Statement

Using Euler's formula, prove that a planar graph has a vertex of degree at most 5.

#### Solution

Let  $G$  be a simple planar graph with  $v \geq 3$  vertices and  $e$  edges. By Euler's formula,

$$v - e + f = 2.$$

Since  $G$  is planar and simple, every face has degree at least 3. By double counting edge-face incidences,

$$3f \leq 2e.$$

Substituting into Euler's formula gives

$$v - e + \frac{2e}{3} \geq 2,$$

hence

$$e \leq 3v - 6.$$

By the handshaking lemma,

$$\sum_{v \in V} \deg(v) = 2e \leq 6v - 12.$$

Therefore, the average degree is strictly less than 6, and hence there exists at least one vertex of degree at most 5.

## Question 2

### Statement

We say that a planar graph  $G$  is *6-minimal* if for any vertex  $v$ , the graph  $G \setminus v$  can be colored with 6 colors. Prove the 6-color theorem for planar graphs by considering a 6-minimal planar graph which cannot be colored with fewer than 7 colors.

### Solution

Assume for contradiction that there exists a planar graph  $G$  that requires at least 7 colors. Among all such graphs, choose one with the minimal number of vertices. Then  $G$  is 6-minimal.

By Question 1,  $G$  has a vertex  $v$  of degree at most 5. By minimality, the graph  $G \setminus v$  admits a 6-coloring. Since  $v$  has at most 5 neighbors, these neighbors use at most 5 distinct colors. Hence there exists at least one color among the 6 colors that is not used by any neighbor of  $v$ .

Assign this color to  $v$ . This extends the 6-coloring of  $G \setminus v$  to a 6-coloring of  $G$ , contradicting the assumption that  $G$  requires at least 7 colors.

Thus every planar graph is 6-colorable.

## Question 3

### Statement

1. Construct a planar graph without triangles whose chromatic number is 3.
2. Find a planar graph different from  $K_4$  which can be colored with 4 colors.

## Solution

(1) An example is any odd cycle, such as  $C_5$ . It is planar, contains no triangles, and its chromatic number is 3.

(2) A standard example is the wheel graph  $W_5$ , obtained by adding a central vertex to a 5-cycle and connecting it to all vertices of the cycle. This graph is planar, is not isomorphic to  $K_4$ , and requires 4 colors.

## Question 4

### Statement

Prove that  $K_8$  cannot be embedded in a torus but can be embedded in a two-holed torus (genus 2).

### Solution

For the complete graph  $K_8$ , we have

$$v = 8, \quad e = \frac{8 \cdot 7}{2} = 28.$$

**Embedding on a torus.** A torus has Euler characteristic 0. If  $K_8$  were cellularly embedded on a torus, then

$$v - e + f = 0 \Rightarrow f = 20.$$

Since  $K_8$  is simple, every face has degree at least 3, hence

$$3f \leq 2e \Rightarrow 3 \cdot 20 \leq 56,$$

which is false. Therefore  $K_8$  cannot be embedded on a torus.

**Embedding on a genus 2 surface.** For an orientable surface of genus 2,

$$\chi = 2 - 2g = -2.$$

Thus,

$$8 - 28 + f = -2 \Rightarrow f = 18.$$

Now

$$3f = 54 \leq 56 = 2e,$$

so there is no combinatorial obstruction. Indeed, it is known that  $K_8$  admits a cellular embedding on a surface of genus 2.



## Question 5

### Statement

Prove that any graph cellularly embedded in a torus has a vertex of degree at most 6.

### Solution

Let  $G$  be a graph cellularly embedded on a torus. The Euler characteristic of the torus is 0, so

$$v - e + f = 0.$$

As before, every face has degree at least 3, so

$$3f \leq 2e.$$

Substituting  $f = e - v$  into this inequality gives

$$3(e - v) \leq 2e \Rightarrow e \leq 3v.$$

Using the handshaking lemma,

$$\sum_{v \in V} \deg(v) = 2e \leq 6v.$$

Thus the average degree is at most 6, and hence there exists a vertex of degree at most 6.

This completes the proof of the 6-degree theorem for graphs embedded on the torus.

## Part 4 – Color’s Theorem

### 1 Surface Identification (Question 1)

To determine the topology of the maps in Figure 2, we analyze the boundary identifications by observing the continuity of the colored regions.

- **Torus (Figure 2-a):** The map is periodic in both horizontal and vertical directions. A color exiting the left boundary reappears on the right at the same height, and colors exiting the top reappear at the bottom. This satisfies the  $a_1 b_1 a_1^{-1} b_1^{-1}$  identification.
- **Annulus (Figure 2-c):** The map is periodic horizontally (left-right match) but has two distinct boundaries at the top and bottom.
- **Möbius Band (Figure 2-b):** The map is periodic in one direction but with a vertical flip (twist), corresponding to the non-orientable scheme  $a_1 a_1$ .
- **Disk (Figure 2-d):** No boundary colors match opposite sides; all four edges are distinct boundaries.

## 2 Euler Characteristic of the Torus Map (Question 2)

Let  $G^* = (V^*, E^*, F^*)$  be a map cellularly embedded on a torus where every vertex has degree 3.

*Proof.* Since  $G^*$  is 3-regular, the handshaking lemma gives:

$$3|V^*| = 2|E^*| \implies |E^*| = \frac{3}{2}|V^*| \text{ [cite: 54]} \quad (1)$$

The Euler characteristic  $\chi(S)$  for a torus ( $g = 1$ ) is:

$$\chi(S) = |V^*| - |E^*| + |F^*| = 2 - 2(1) = 0 \text{ [cite: 54]} \quad (2)$$

Substituting the edge relation into the Euler formula:

$$|V^*| - \frac{3}{2}|V^*| + |F^*| = 0 \implies -\frac{1}{2}|V^*| + |F^*| = 0 \implies |V^*| = 2|F^*| \quad (3)$$

Consequently,  $|E^*| = \frac{3}{2}(2|F^*|) = 3|F^*|$ . □

## 3 Dual Graph and Chromatic Number (Question 3)

### Statement

We associate a graph  $G = (V, E)$  to the map  $G^*$ . Provide an explicit embedding of  $G$  in the torus model and an abstract representation of  $G$  with all vertices on a circle. What is this graph, and what is its chromatic number  $C(G)$ ?

### Solution

The graph  $G$  is the dual graph of the map  $G^*$ . Since each face of  $G^*$  corresponds to a vertex of  $G$ , and each vertex of  $G^*$  has degree 3, every face of  $G^*$  is adjacent to three others.

Hence  $G$  is a 6-vertex complete graph  $K_6$  embedded on the torus. An explicit embedding is obtained by placing the vertices in the square torus model and routing edges using the identifications.

The chromatic number of  $K_6$  is

$$C(G) = 6.$$

## 4 Optimal Color's Theorem for the Torus (Question 4)

*Proof.* The Heawood formula for the chromatic number  $\chi(S)$  of a surface of genus  $g$  is:

$$H(g) = \left\lfloor \frac{7 + \sqrt{1 + 48g}}{2} \right\rfloor \quad (4)$$

For a torus,  $g = 1$ :

$$H(1) = \left\lfloor \frac{7 + \sqrt{1 + 48(1)}}{2} \right\rfloor = \left\lfloor \frac{7 + 7}{2} \right\rfloor = 7 \quad (5)$$

Since we have explicitly shown that  $K_7$  can be embedded in the torus (Question 3), the bound is tight. Thus,  $C(G) = 7$  is optimal.  $\square$

## 5 Two-Holed Torus Conjecture (Question 5)

**Conjecture:** The chromatic number for a two-holed torus ( $g = 2$ ) is 8.

*Proof.* Applying the Heawood formula for  $g = 2$ :

$$H(2) = \left\lfloor \frac{7 + \sqrt{1 + 48(2)}}{2} \right\rfloor = \left\lfloor \frac{7 + \sqrt{97}}{2} \right\rfloor \approx \lfloor 8.42 \rfloor = 8 \quad (6)$$

To prove this is reachable, we check if  $K_8$  can be embedded in  $g = 2$ . From the genus formula for complete graphs:

$$g(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil \quad (7)$$

For  $n = 8$ :

$$g(K_8) \geq \left\lceil \frac{(8-3)(8-4)}{12} \right\rceil = \left\lceil \frac{20}{12} \right\rceil = 2 \quad (8)$$

Since  $g(K_8) = 2$ ,  $K_8$  can be embedded in a two-holed torus, making 8 colors necessary and sufficient.  $\square$

### Edge-maximal planar graphs.

- Let  $e$  be an edge in an edge-maximal graph embedded in the plane such that every triangle containing  $e$  is a face of the embedding. Prove that an edge-maximal graph is still edge maximal after contracting an edge.

**Answer.** Let  $V, E$  be the number of vertices and edges respectively of an edge-maximal planar graph  $G$ . When contracting an edge  $e$ , the decrease in the number of edges is one more than the number of triangles containing that edge. If  $e$  satisfies the condition above, contracting  $e$  forms a minor  $G'$  with  $V' = V - 1$  vertices and  $E' = E - 3$  edges. Planar graphs are edge maximal if and only if  $E = 3V - 6$ . The new graph  $G'$  has

$$E' = E - 3 = 3V - 9 = 3V' - 6,$$

and so it is also edge-maximal.

- Prove that plane triangulations with at least 4 vertices are 3-connected.

**Answer.** Fix an embedding of the triangulation. First, we prove that every plane triangulation with  $n \geq 4$  vertices has at least  $n$  edges  $e$  with the property that the only triangles containing  $e$  are faces.

The proof is by induction on  $n$ . The base case is  $K_4$  and holds trivially because all 6 edges have this property. For the induction, pick any triangle  $t$  that is not a face. We split the graph into two parts,  $G_1 = (V_1, E_1)$  and  $G_2 = (V_2, E_2)$ , where  $G_1$  is the induced graph on  $t$  and all of the vertices inside  $t$  in the embedding, and  $G_2$  is the induced graph on  $t$  and all of the vertices outside  $t$ . Both graphs are plane triangulations and both have fewer than  $n$  vertices. So, by induction, there are  $|V_1| + |V_2| = n + 3$  edges in the two graphs with the desired property. When we combine the graphs, the only edges that can lose this property are the edges of  $t$ , because no edges pass between any of the other vertices of  $G_1$  to any of the other vertices of  $G_2$ . Thus, at most 3 edges with the desired property are lost in the combining, leaving us with

$$|V_1| + |V_2| - 3 = n$$

such edges, as desired.

Now we can prove that plane triangulations with at least 4 vertices are 3-connected. Suppose for contradiction that  $G$  is not 3-connected. Then there is a pair of vertices  $u, v$  that disconnect it. Let  $X$  and  $Y$  be two disconnected induced subgraphs of  $G \setminus \{u, v\}$ . By the preceding claim, as long as there are at least 4 vertices, we can contract an edge. Repeatedly contract edges, keeping the graph edge-maximal, until only 4 vertices remain. We must be careful never to contract an edge between  $u$  and  $v$  and to keep around at least one vertex of  $X$  and at least one vertex of  $Y$ . The resulting graph is edge-maximal with 4 vertices and therefore must be  $K_4$ . However, the vertices  $u, v$  still disconnect the graph. Since  $K_4$  is 3-connected, this is a contradiction.